Design, Simulation, and Sound-Card Characterization of First-Order RC Filters

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***Abstract*—**This project designed, simulated, constructed, and experimentally characterized first-order RC low-pass and high-pass filters. A nominal 1.5 kΩ resistor and 0.1 µF capacitor produced a calculated cutoff frequency of 1.061 kHz. MATLAB was used to calculate the ideal magnitude response and to automate sinusoidal measurements, while LTspice verified the expected magnitude and phase behavior. Physical breadboard circuits were driven and measured through the motherboard-integrated Realtek analog audio interface of an Alienware Aurora R7 using line-out and line-in connections. The measured responses followed the expected first-order behavior, with mean absolute magnitude errors of approximately 0.55 dB for the low-pass filter and 0.59 dB for the high-pass filter. Direct DMM measurements of 1475 Ω and 100.3 nF yielded an adjusted cutoff prediction of 1.076 kHz, indicating that most remaining error arose from the measurement chain and other nonideal effects rather than component tolerance alone.

Keywords*—RC filter, low-pass filter, high-pass filter, frequency response, MATLAB, LTspice, sound card, experimental validation.*

I. INTRODUCTION

First-order RC filters provide a direct connection between circuit analysis and frequency-domain behavior. The objective of this project was to complete a full design-and-test cycle for complementary low-pass and high-pass filters using theoretical calculations, MATLAB modeling [1], LTspice simulation [2], breadboard construction, and experimental measurements. The same nominal resistor and capacitor values were used for both configurations so that the cutoff frequency remained common while the passband and stopband behavior reversed.

II. THEORY AND DESIGN

The nominal component values were R = 1.5 kΩ and C = 0.1 µF. The cutoff frequency for a first-order RC filter is given by (1).

$$f_c = \frac{1}{2\pi RC} \tag{1}$$

Using the nominal values, (1) gives f_c = 1.061 kHz. The low-pass output was taken across the capacitor and the high-pass output across the resistor. Their continuous-time transfer functions are shown in (2) and (3).

$$H_{\mathrm{LP}}(j\omega) = \frac{1}{1 + j\omega RC} \tag{2}$$

$$H_{\mathrm{HP}}(j\omega) = \frac{j\omega RC}{1 + j\omega RC} \tag{3}$$

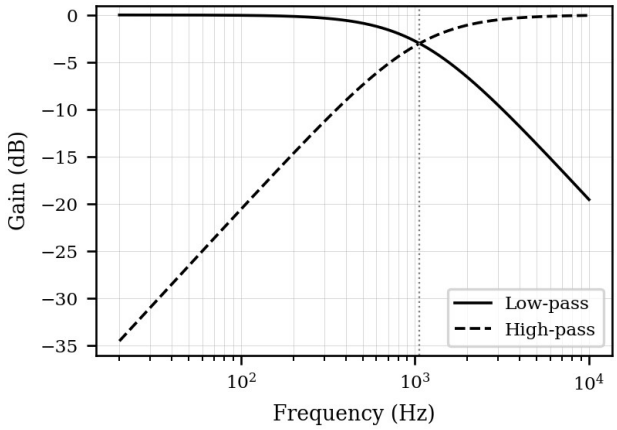


Fig. 1. Theoretical magnitude responses using the nominal component values.

III. SIMULATION AND EXPERIMENTAL METHOD

A. LTspice Simulation

Both filter configurations were simulated with a 1 V small-signal AC source using a 20 Hz to 10 kHz decade sweep at 100 points per decade. The schematics used the same nominal component values as the MATLAB model. The simulated cutoff region occurred near 1.06 kHz, where the magnitude was approximately -3 dB. The simulated low-pass phase was approximately -45° at cutoff and the high-pass phase approximately +45°.

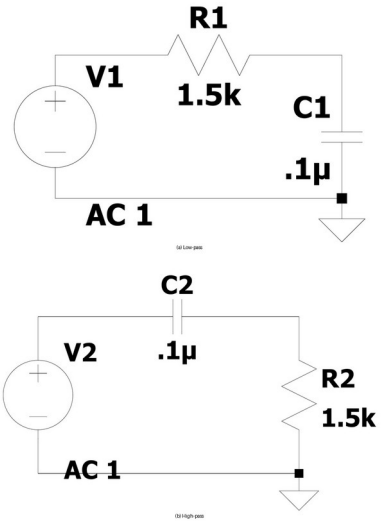


Fig. 2. LTspice schematics: (a) low-pass and (b) high-pass configurations. The AC sweep was 20 Hz to 10 kHz at 100 points/decade.

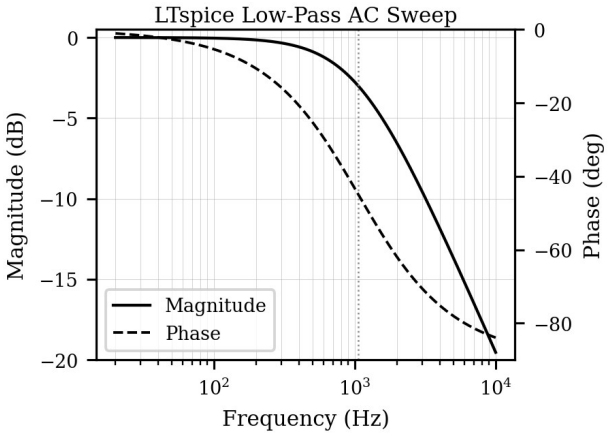


Fig. 3. Publication-scale redraw of the LTspice low-pass AC sweep, showing magnitude and phase from 20 Hz to 10 kHz.

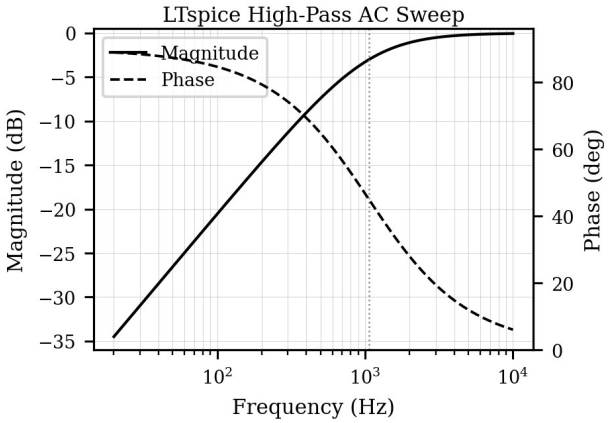


Fig. 4. Publication-scale redraw of the LTspice high-pass AC sweep, showing magnitude and phase from 20 Hz to 10 kHz.

B. Audio Measurement Method

The filters were assembled on a solderless breadboard. MATLAB generated 2 s sine waves at a 48 kHz sampling rate with a normalized amplitude of 0.05. The test frequencies were 100, 250, 500, 1061, 2000, 5000, and 10000 Hz. The analog line-out of the Alienware Aurora R7 motherboard sound card drove the circuit, and the motherboard Realtek line-in recorded the response. Windows playback volume was held at 10/100 and the line-in recording level at 80/100 for all runs.

At each frequency, a direct bypass measurement was first recorded to establish the reference amplitude. The filter was then inserted without changing software or operating-system levels. RMS amplitude was calculated over the steady-state interval from 0.7 s to 1.3 s according to (4), and filter gain was calculated from the ratio of filtered to reference RMS amplitude using (5).

$$V_{\mathrm{rms}} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} v_n^2} \tag{4}$$

$$G_{\mathrm{dB}} = 20\log_{10}\left(\frac{V_{\mathrm{out,rms}}}{V_{\mathrm{in,rms}}}\right) \tag{5}$$

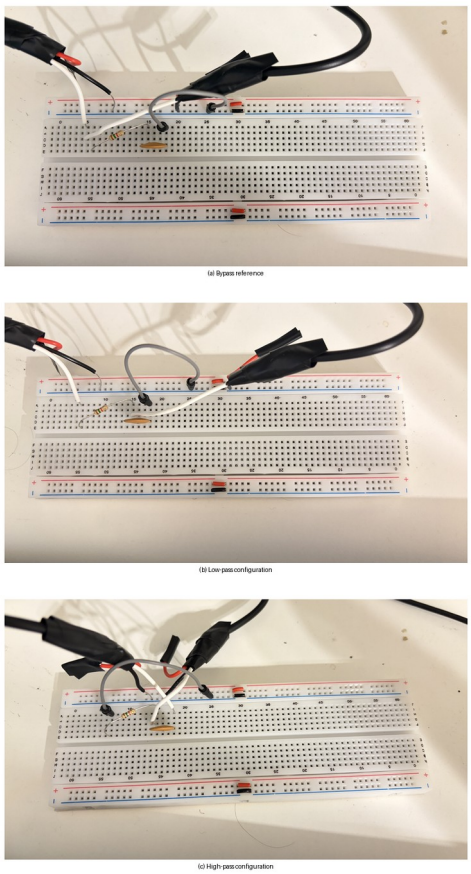


Fig. 5. Experimental wiring: (a) bypass reference, (b) low-pass filter, and (c) high-pass filter.

IV. RESULTS

The measured low-pass and high-pass responses followed the theoretical curves across the full test range. At the nominal cutoff frequency of 1061 Hz, the low-pass measurement was -3.665 dB and the high-pass measurement was -3.773 dB, compared with ideal values of approximately -3.01 dB.

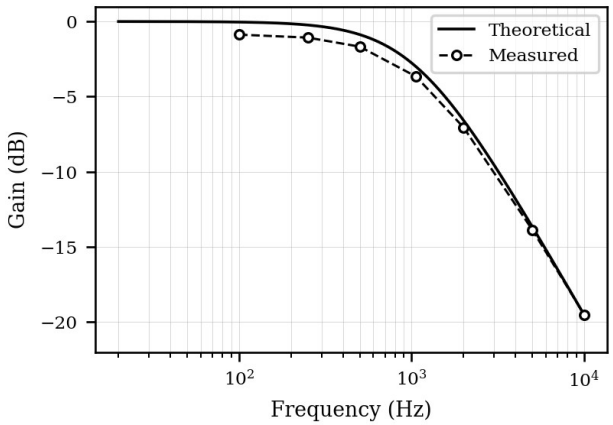


Fig. 6. Low-pass theoretical and experimental magnitude response.

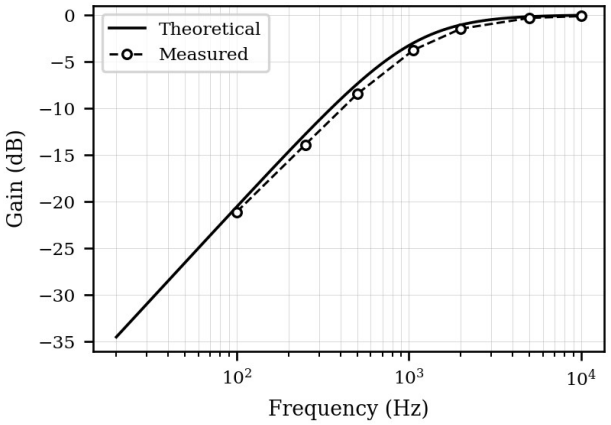


Fig. 7. High-pass theoretical and experimental magnitude response.

TABLE I. THEORETICAL AND MEASURED GAIN (DB)

f (Hz)	LP Theory	LP Meas.	HP Theory	HP Meas.
100	-0.04	-0.88	-20.55	-21.10
250	-0.23	-1.07	-12.79	-13.89
500	-0.87	-1.67	-7.41	-8.48
1061	-3.01	-3.67	-3.01	-3.77
2000	-6.58	-7.05	-1.08	-1.49
5000	-13.66	-13.86	-0.19	-0.34
10000	-19.53	-19.51	-0.05	-0.14

V. COMPONENT VERIFICATION AND DISCUSSION

After the frequency-response measurements, the resistor and capacitor were measured directly with a CEN-TECH CM300 digital multimeter. The resistor display read 1475 Ω and the capacitor display read 100.3 nF, as shown in Fig. 8. Substitution of these measured values into (1) gives an adjusted cutoff frequency of approximately 1.076 kHz, only about 1.4% above the nominal prediction. This shift is too small to explain the full measured magnitude offset by itself.

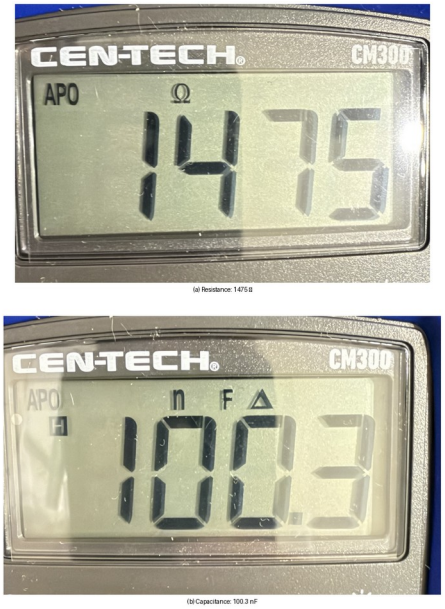


Fig. 8. Direct component measurements: (a) resistor, 1475 Ω; (b) capacitor, 100.3 nF.

The mean absolute magnitude error was 0.55 dB for the low-pass filter and 0.59 dB for the high-pass filter. The low-pass error decreased from about 0.84 dB at 100–250 Hz to nearly zero at 10 kHz. The high-pass error was largest around 250–500 Hz and decreased toward higher frequencies. Because the component values were close to nominal, the remaining discrepancy is more consistent with nonidealities in the complete measurement chain, including finite sound-card source and input impedances, frequency-dependent audio-interface response, breadboard and contact resistance, and RMS measurement uncertainty.

The sound-card method provided a low-cost way to obtain a useful frequency-response measurement without a bench function generator or oscilloscope, but it should not be treated as an ideal instrumentation system. A future extension could independently characterize the Realtek audio path, use simultaneous two-channel acquisition, or repeat the experiment with laboratory instruments for comparison.

VI. CONCLUSION

The project successfully demonstrated the design, simulation, construction, and measurement of first-order RC low-pass and high-pass filters. MATLAB and LTspice predicted a nominal cutoff near 1.061 kHz, and the physical filters reproduced the expected first-order roll-off and complementary passband behavior. Direct component measurements predicted a cutoff of approximately 1.076 kHz, confirming that component tolerance was a relatively small contributor to the experimental deviations. The final results connect ideal circuit theory to simulation and physical measurement while illustrating the influence of a nonideal audio measurement interface.

REFERENCES

[1] The MathWorks, Inc., MATLAB, Natick, MA, USA.
[2] Analog Devices, Inc., LTspice, Wilmington, MA, USA.